

Learning Designs and Learning Analytics

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ABSTRACT

Government and institutionally-driven reforms focused on quality teaching and learning in universities emphasize the importance of developing replicable, scalable teaching approaches that can be evaluated. In this context, learning design and learning analytics are two fields of research that may help university teachers design quality learning experiences for their students, evaluate how students are learning within that intended learning context and support personalized learning experiences for students. Learning Designs are ways of describing an educational experience such that it can be applied across a range of disciplinary contexts. Learning analytics offers new approaches to investigating the data associated with a learner's experience. This paper explores the relationship between learning designs and learning analytics.

Categories and Subject Descriptors

H.5 Information Interfaces and Presentation

General Terms

Measurement, Design, Human Factors

Keywords

learning design, pedagogical models, learning analytics, social network analysis, university teaching.

1. INTRODUCTION

Within the higher education sector internationally there is mounting interest in developing replicable and scalable exemplars and templates of effective teaching practice. This, in part, has arisen from increased accountability and quality assurance expectations in terms of the demonstration of teaching quality. Bodies such as the Quality Assurance Agency for Higher Education (QAA) in the UK and the Australian Universities Quality Agency (AUQA) in Australia have been charged assessing and reporting on the quality of the education and student services that are provided by the higher education institutions in their countries.

Whether driven by national accountability expectations or institutional initiatives to support quality, the field of learning design has received much attention in the education sphere. This

interest may be due to the perceived potential for learning designs to address concerns about quality. Learning designs, (or as they may be termed pedagogical models or pedagogical patterns) are representations of teaching practice. Learning designs articulate a sequence of learning activities, what resources are used by learners engaged in those activities, and how the teacher or others participants in the experience supports the learning activities. A learning design can act as a model or template that teachers may replicate across a range of educational contexts. In essence, learning designs help frame the intent of and process for the pedagogical experience.

Learning design research and its up-take in educational practice has come in many forms. At its basis is the concept that it is important to describe an educational experience in such a way that it can be replicated then applied across a range of disciplinary contexts. Essentially, the focus is on the design and planning of the educational experience. In practice, learning designs are predominately directed at the designer/teacher of the educational experience. Learning designs provide a way for a teacher to document their pedagogical intent. When shared amongst teachers, learning designs provide stimulus or guidance for other teachers to design quality educational experiences.

Learning analytics, on the other hand, are largely directed at the student. In essence learning analytics allow us to use the data associated with a learner's interactions to make pedagogically informed decisions and evaluations. Learning analytics as an emerging interdisciplinary field draws upon and integrates research related to data mining, social networks, data visualisation, machine learning, semantics, e-learning and educational theory and practice. Seemingly, learning analytics takes up where learning design finishes in the educational experience continuum – implementation and outcomes.

This paper explores the relationship between learning design and learning analytic approaches. The paper considers the context in which learning design research emerged and highlights one strand of its development in Australia as a way of consider how it may be compatible with learning analytics. It also considers work in social network analysis as an example subset of learning analytics to demonstrate how this field complements learning design.

2. BACKGROUND

Although the terminology and the scope of research and application of both learning design and learning analytics approaches have evolved in recent years, their underlying basis is not new. Essentially these research fields are concerned with documenting and reporting on educational practice and experience. The interest in both learning designs and learning analytics has gained momentum within the context of two compatible worldwide trends within the higher education environment – calls for improved teaching and increasing use of

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Soon after this focus on quality teaching began, the increasingly accessible and user-friendly Web provided universities with an opportunity to expand their student markets and offer flexible forms of programs. There was increasing expectations for, and interest among, university teachers to integrate technology within their teaching practice. From those early days of the Web, research and teaching scholarship explored the opportunities that technology provided education – particularly in terms of improving pedagogy [2]. The increasing emphasis on and expectations towards technology integration created complexity within the teaching environment in terms of its design, implementation and evaluation process. As a result, professional development and teaching innovation funding was directed at projects aimed at improving teaching using technology. Underlying the support for such projects was the premise that the project outcomes, resources and products would be disseminated amongst the higher education community and adopted at the institutional and individual teacher levels. The emergence and subsequent developments in the area of learning designs exemplify these principles of resource creation, sharing and reuse that were at the heart of this higher education funding.

For over a decade now, researchers around the world, have been interested in the educational and technical issues associated with how teachers and instructional designers could create document, share, adapt and implement, educational design ideas. This involved developing a technical language (IMS LD [3]), developing tools in which learning designs could be created and/or used [4, 5], and investigating how teachers interpret and use different learning design representations [6, 7].



The exemplar and generic learning designs developed through this Australian project have been the basis of a number of subsequent research studies. The learning design framework has been found to be useful as method for teachers to interpret and then apply the ideas in the design of their own teaching [10] and as a tool that that allows extraction of teaching and learning processes for analysis [11]. The learning design framework has proven to be useful for teacher interested in how to integrate learning objects into their teaching – where learning objects are consider the ‘resources’ within the learning designs [12]. In researching their dissemination, these learning designs have been reportedly used by staff in educational development units and individual university teachers to support the process for designing courses and also as a way of reflecting on the implementation once a course has been delivered [13].

This combined work suggests that a learning design framework of some sort is a useful mechanism to support the design and planning aspect of the educational experience and also as a way of sharing and stimulating educational ideas amongst university teachers. Thus, this research basis gives us a range of mechanism to define pedagogical intent. The next step in the process is being able to understand the process and outcome of these intentions. While traditional course assessment provides one mechanism, the

growing availability of data through online education systems allows other opportunities.

3. THE POTENTIAL OF LEARNING ANALYTICS

While learning designs may provide theoretical, practice-based, and/or evidence-based examples of sound educational design, learning analytics allow us to test those assumptions with actual student interaction data in lieu of self-report measures such as post hoc surveys. In particular, learning analytics provides us with the necessary data and tools to support the accountability that has been called for in higher education.

As with learning designs, learning analytics takes a range of forms and foci. One sub-set of the learning analytics domain is social network analysis (SNA). SNA provides a way to understand the educational experience and outcomes for learners engaged in online communication activities. Much of the aforementioned take-up of technology in higher education has been the associated with the the discussion forum – particularly within learning management system course sites [15, 16]. There are countless examples in the literature that describe the design and implementation of educational programs that use discussion forums. Where these cases are aligned with a theoretical premise, it is of the social cognitive aspect of learning and the benefits of collaboration (e.g., [17]). Thus it follows that, social network analysis provides a methodology for teachers to begin to understand the impact of their implemented learning design on the student learning experience and outcomes. However, this analysis needs to be readily accessible and interpretable for all educators regardless of expertise in network analyses; information and communication technologies or educational theory.

To address this issue of user uptake and interpretation, Dawson [18] and colleagues developed an analytical tool named Social Networks Adapting Pedagogical Practice (SNAPP – available at <http://research.uow.edu.au/learningnetworks/seeing/snapp/index.html>). SNAPP seamlessly integrates with learning management system discussion forums to extract data and provide teachers with real-time visualizations of the discussion forum activity (see figure 2). Recent evaluations of the use of SNAPP by university teachers demonstrated that the tool was seen to be particularly effective for retrospective analysis of an education implementation after a course was completed [19]. However, despite the ease in generating ongoing visualizations of student discussion interactions and relationship development, teachers did not take the opportunity to analyze data during actual course implementation. This process would have allowed for modification of the learning design ‘on the fly’ if the intended experience or outcomes were not being realized. While this learning analytics tool theoretically supports the implementation stage of that educational continuum - to date it has been largely used as a reflective tool. Further research is required to investigate how educators can be better prompted when student interactions indicate a deviation from the intended learning outcomes. For example, an educator observes minimal student-to-student interactions arising despite the implementation of learning activities designed to establish a learning community.

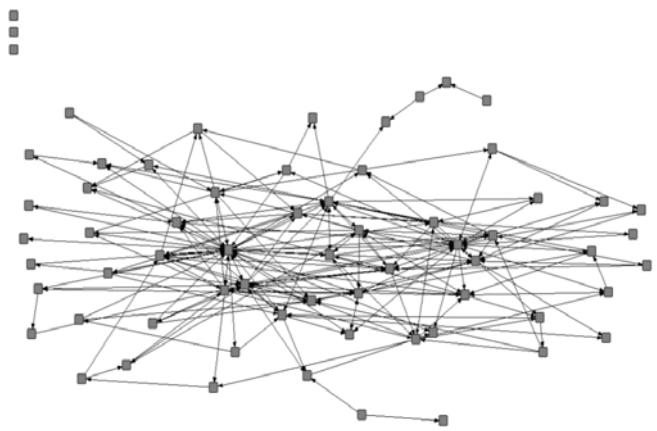


Figure 2. SNAPP generated sociogram of the student network relationships evolving from an online discussion forum (student names have been removed).

4. THE CHALLENGE

Essentially, the learning design approach is founded on a case-based reasoning premise. A documented learning design provides a case that is abstract enough such that teachers can imagine how it might be applied to their own context. However, investigation of the use of learning designs suggests that teachers also benefit from understanding the context of others and the environments in which they teach and thus cases should not be so abstract as to lose a connection with the original context [20]. Regardless of the level of abstraction, the notion is that teachers learn or refine their ideas about design from the examples of other teachers.

The dissemination and adoption of innovative teaching ideas in higher education – whether those ideas are expressed through learning design representations or other types of tools or resources, recognizes the social processes of being a teacher. University teachers report that their ideas about teaching come from interacting with colleagues and other teachers [21, 22]. In this context, teachers are less likely to seek information from external resources or online sources. This research (and that of others investigating educational change) frequently highlights time as a barrier to the adoption of new ideas into ones teaching practice. Academics in higher education institutions are stretched across their teaching and research responsibilities. They are more likely to invest time to try new ideas if they are convinced, through evidence, that the innovation will have an impact for them and their learners that is both time and workload efficient.

Learning analytics has the potential to provide this evidence. The research challenge is identifying effective and efficient ways to make learning design and learning analytics useful and relevant for teachers. The integration of research related to both learning design and learning analytics provides the necessary contextual overlay to better understand observed student behavior and provide the necessary pedagogical recommendations where learning behavior deviates from pedagogical intention.

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